

1

→ Intelligent behaviour of a bio-inspired artificial agent:

- ↳ Make decisions
- ↳ Reach to something new
- ↳ Learn and develop
- ↳ Interpret images and scenes
- ↳ Communicate and cooperate

→ Requirements: → Consider findings about intelligent behaviour in nature
→ learn, represent and process based on bio-inspired principles

(Navigation)

Eg: place cells (inner map of Env), grid cells (form coordinate system for Nav)

- ↳ active in a particular place
- ↳ reaching for passing certain locations

↳ both combined: comprehensive inner positioning system

⇒ Bio-inspired AI allows to solve problems informed by the best problem solvers in nature

2

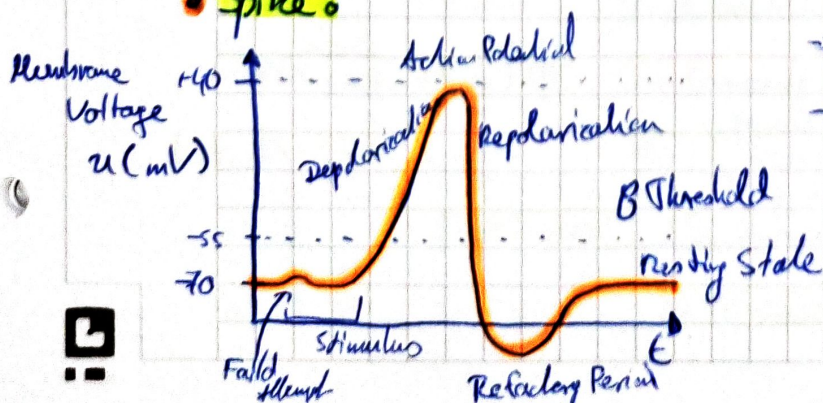
Spiking Neural Networks

- Advantages Nervous System:
 - 1) Selective transmission of signals across distant areas = more complex bodies and brains
 - 2) Complex adaption to environment changes

• What makes brains different:

- Components & behavior of neurons ^{very} similar across animal species
- Development in brain how neurons are interconnected

• Spike:



→ this cycle ~ 3-50 ms

→ once over threshold then full spike!

Stimuli arriving at dendrites are transferred via axon to synapses

↳ activity measured as spikes

activity axon → squid

L2

- Types of neurons:
 - 1) Excitatory
 - 2) Inhibitory

- Modelling Neuron Communication:

→ Weight Matrix:

Coding based on activity level

$\geq \theta$ → 1
else → 0

Mc Culloch - Pits / Perceptron

"rate-coded"

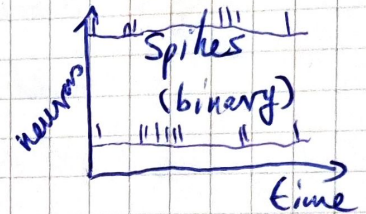
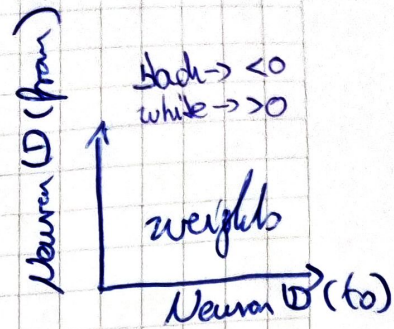
~~Connectionism~~ Connectionism

→ Binary Matrix Models:

Coding based on activity frequency / timing

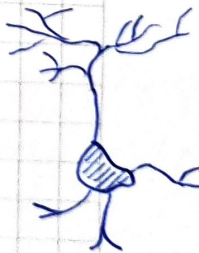
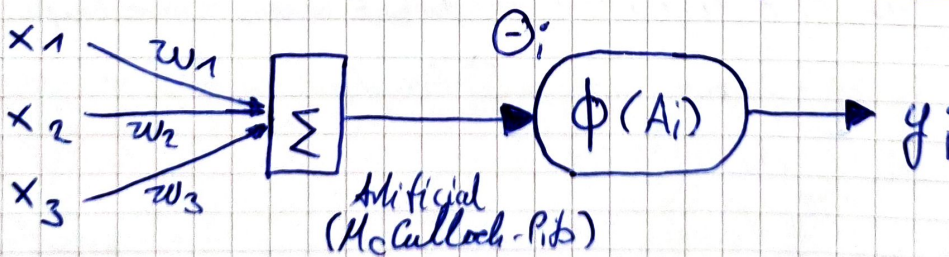
Spiking Neurons

Computational Neuroscience



- Artificial Rate-Coded Neuron Unit:

$$y_i = \phi(A_i) = \phi\left(\sum_j w_{ij} x_j - \theta_i\right)$$



eg: Identity, Step, Sigmoid function

Sigmoid - function:

continuous,
non-linear,
monotonic,
bounded,
asymptotic

$$\phi(A_i) = \frac{1}{1 + e^{-k A_i}}$$

$$= \tanh(h A_i)$$

L2

Threshold to Bias Unit

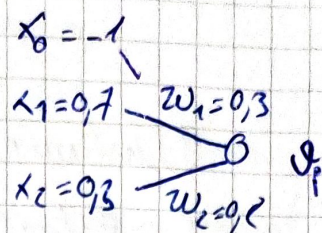
instead of threshold θ_i

↳ additional weighted input x_0 : always -1.0 + weight w_{i0}

$$y_i = \phi(x_i) = \phi\left(\sum_{j=0}^n w_{ij} x_j - \theta_i\right)$$

$$\rightarrow y_i = \phi(x_i) = \phi\left(\sum_{j=0}^n w_{ij} x_j\right)$$

→ easier to express / program



How can a Neural Unit represent Similarity?

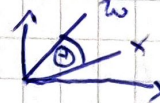
Output of a neuron as a measure of similarity between its input pattern and its pattern of connection weights

1) Linear Algebra: $y = \phi\left(\sum_i w_i x_i\right)$, $y = w \cdot x$

2) Similarity/Angle between two vectors:

$$\cos \theta = \frac{w \cdot x}{|w||x|}, \quad 0 \leq \theta \leq \pi, \quad |x| = \sqrt{x \cdot x}$$

3) Output Signal: $w \cdot x = |w||x| \cos \theta$

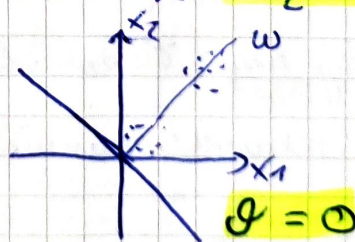
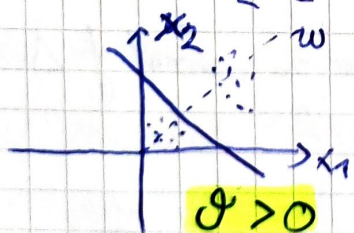


How can a Neural Unit separate Input Patterns?

A neuron divides the input space in two regions, one where $A \geq 0$ and one where $A < 0$:

$$w_1 x_1 + w_2 x_2 - \theta = 0 \Leftrightarrow$$

$$x_2 = \frac{\theta}{w_2} - \frac{w_1}{w_2} x_1$$



How can we represent Input Vector Encoding

A	B	C
●	○	○
○	●	○
○	○	●

Local:

One neuron stands for one item, one another cell,

Scalability problem, Robustness problem,

use this in practice with Softmax function

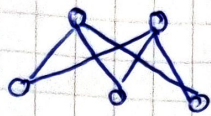
A	B	C
●	●	○
○	●	●
●	○	●

Distributed: Neurons encode features, > 1 item, one item activated > 1 neuron

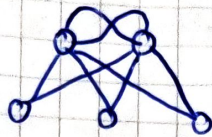
Better Scalability, Robust to damage or noise

L2

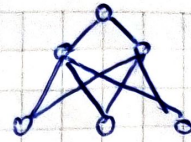
Architectures:



Feed-forward



recurrent

feed-forward
multi-layerfully
connected

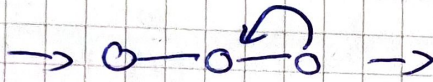
Dynamic Neural Models:

→ Rate-coded model is rather static

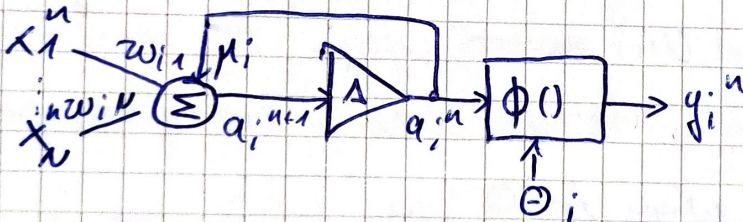
Before: Output is determined by current input, no temporal processingNow: Temporal Dependencies (eg: Speech / Language)

Realize in Static Networks:

feedback loop between output from time t to act input
at time $t+1$ provides temporal delay of signal into the future



Discrete Dynamic Neural Model:

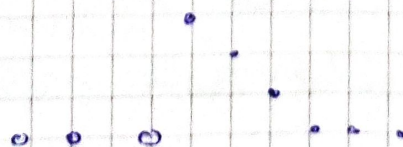
 Δ : Time delay between current activation $a_i(n)$ and update step $a_i(n+1)$ Delay keeps the activation a_i for an amount of Δ μ_i : recurrent connection weight

→ update:

$$a_i(n+1) = \mu_i a_i(n) + \sum_{j=1}^N w_{ji} x_j(n)$$

→ output:

$$y_i(n) = \phi(a_i(n) - \theta_i)$$

Activation Decay for different μ $\mu = 0, 1$ 

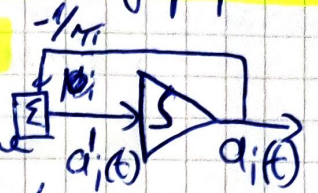
L2

Continuous Model:

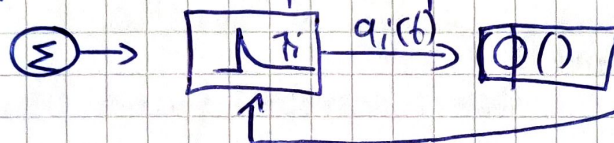
Derivation of activation over time t :

$$\frac{d a_i(t)}{dt} = -\frac{1}{\tau} a_i(t) + \sum_{j=1}^N w_{ij} x_j(t)$$

- Displays exponential dynamics of activation at neuron's membrane
- τ is a time constant whose magnitude is inversely proportional to the decay rate of the activation for $\tau > 0$
- update is on derivative of activation over time
- ↳ $\mu \approx$ exponential decay e^{-t/τ_i}



→ Simplification as Compact Representation



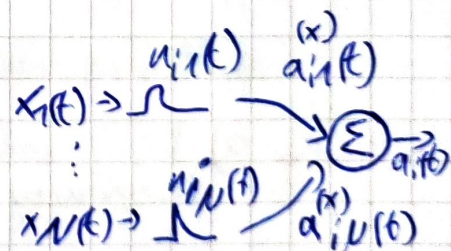
μ (discrete case)
↓
 τ (continuous case)

→ Simplification of so-called Hodgkin and Huxley Model

- ↳ Neuron Model: Voltage difference of a capacitor
- ↳ τ is approximated by a resistor

Neuron Model with different Input Dynamics

each synapse contributes with different temporal characteristics



→ u_{ij} : synaptic kernel = temporal characteristic for input

→ temporal convolution of input x with different kernels u_{ij}

Bio-inspired AI

19.01.2024

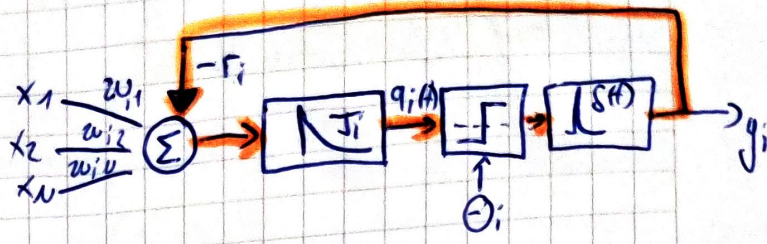
L2



Dynamic Models:

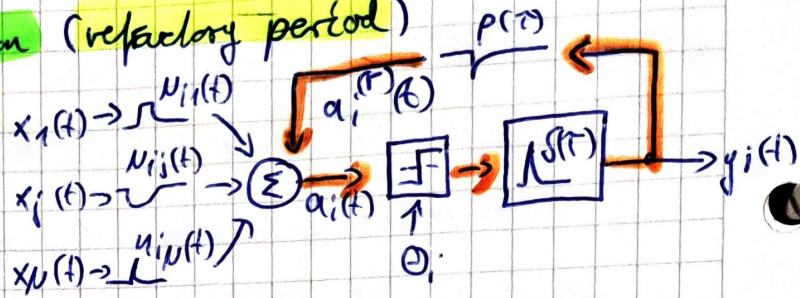
$\tau \frac{d}{dt} u_i = -q_i + R \Sigma(t) // \text{Integration}$

$u_i(t) = 0 // \text{Fire \& Reset // threshold}$



→ Strong negative feedback $-r_i$ after spike, to prevent continuous spike emission (refractory period)

→ Spike Response Model:



→ Inhibitory feedback modeled by exponential function, bringing neuron activation into large negative value after spike (refractory period)

→ Both: → Spikes are function of time: Dirac Impulse $\delta(\tau)$

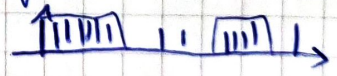
→ Differs: Inhibitory Feedback

Different Spike Encoding Schemes

Spikes represent presence or absence of stimulus: Binary decision

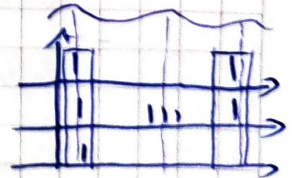
Pow Pow Pow

a) Firing rate represents strength of stimulus (Frequency Code)



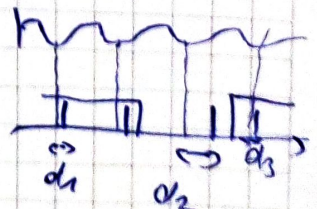
TOGETHER

b) Number of neurons firing together represent stimulus intensity (Temporal Coincidence / Synchronicity)



Stimulus
↓ delay
Fire

c) Encode strength of stimulus depending on the firing delay of the neuron (Delay Code, neurons that receive stronger stimulation fire earlier)



• How Neurons Learn. Hebb's rule:

Cells that fire together, wire together

i.e. activation correlations strengthen synaptic connections

→ constitute the brain plasticity

→ Plasticity: measure for complex task solving capabilities

↳ Synaptic Plasticity: ability of synapses to strengthen or weaken over time

→ Hebb's rule becomes function of time between pre- and postsynaptic neuron

→ Initial weight $\in [0, 1]$

→ Training is described by repeated presentation of input (training) patterns
↳ until training criterion is met

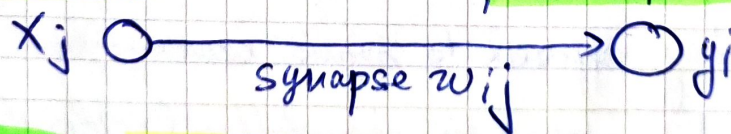
→ Modification of weights after each presentation step (learning)

• Synaptic Plasticity in Hebb's rule

→ Learning is experience-dependent modification of connection weights

pre-synaptic neuron

post-synaptic neuron



Hebb's rule: $\Delta w_{ij} = x_j y_i$

Update: $w_{ij}^t = w_{ij}^{t-1} + \eta \Delta w_{ij}$

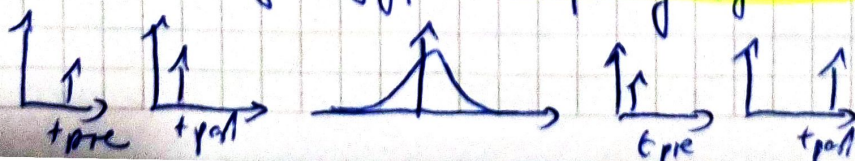
// simplest & most general form of learning

Problems:

→ Hebb did not assume the existence of synaptic weakening

↳ suffers from self-amplification (unbounded growth of weights)

↳ since $\Delta w_{ij} = x_j y_i \rightarrow$ temporally symmetric



L2

• Spike-Timing Dependent Plasticity

→ Hebbian Learning in Spiking NN:

→ Temporal difference between reception of spike from presynaptic neuron and the emission of a spike from the postsynaptic neuron

→ Spike-Timing Dependent Plasticity (STDP)

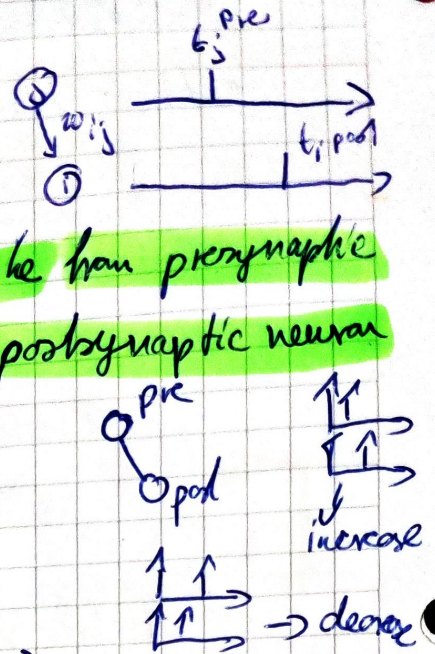
→ Hebbian Learning with Temporal Asymmetry

↳ weights can now decrease

→ Synapse plasticity (weight between neurons)

changes according to STDP function of pre- and postsynaptic spike timings

→ Plasticity increases most when presynaptic neuron fires shortly before postsynaptic neuron



• Summary

→ Spiking NN resemble biological information transmission

↳ activation state of Neuron approximated by firing rate

→ Temporal coding opposed to static learning (input determines output)

→ Learning based on synaptic correlation (Hebb's Rule)

→ Concept of Spike-Timing Dependent Plasticity

→ Different Models: Rate-Coded, Integrate & Fire, Spike Response